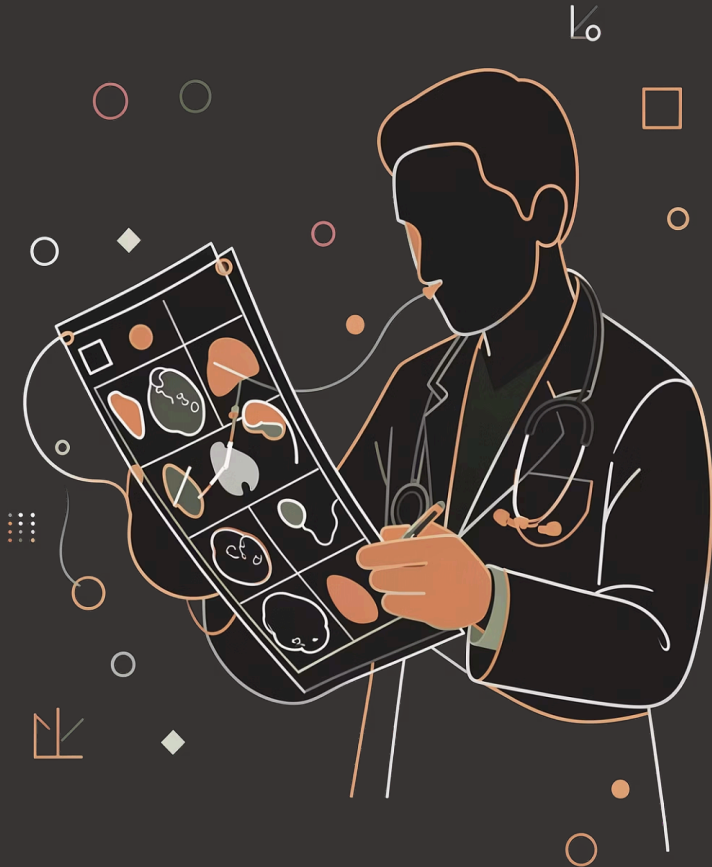




Breast Cancer Diagnosis Prediction using K-Nearest Neighbors (KNN)

By Shafaat Ullah

The Problem: Why Early Diagnosis Matters



Global Burden

Breast cancer is one of the most common cancers worldwide, affecting millions of patients each year.

Survival Impact

Early and accurate diagnosis significantly improves survival rates — timely detection is critical to patient outcomes.

Role of Machine Learning

Machine learning can learn patterns from historical medical data to assist doctors in tumor classification with speed and consistency.

Project Objectives



Analyze the Dataset

Explore the Breast Cancer Wisconsin Diagnostic dataset for structure and statistical patterns.



Train the KNN Classifier

Build and train a KNN model to predict benign vs. malignant tumor classification.



Preprocess Features

Standardize and prepare all features to ensure the model receives clean, normalized input.



Evaluate Performance

Assess model quality using accuracy, classification report, and confusion matrix metrics.



Dataset Overview

Source: Breast Cancer Wisconsin Diagnostic Dataset (Kaggle / UCI)

Features include radius, texture, smoothness, and other cellular measurements. Target variable — **Diagnosis** — is classified as either **Benign (B)** or **Malignant (M)**.

569

Patient Samples

30

Numeric Features

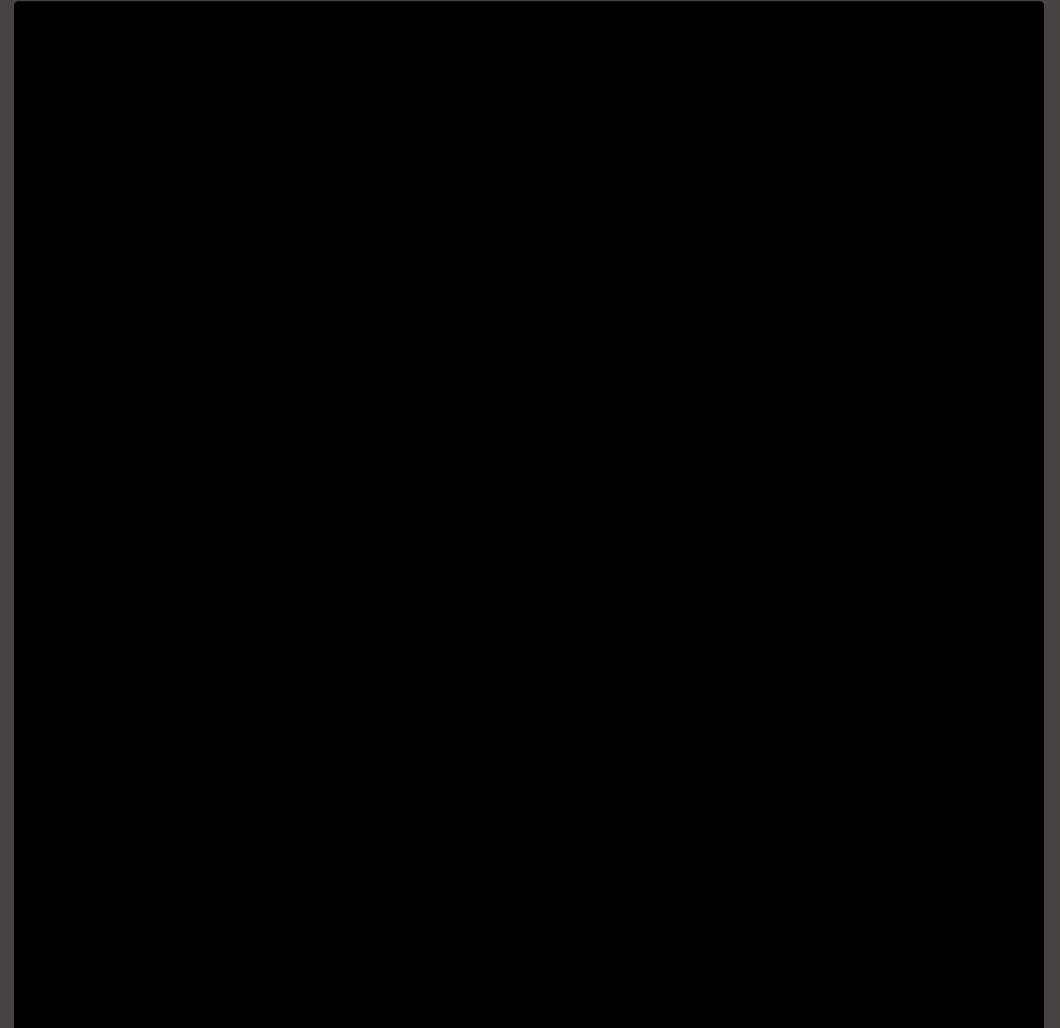
2

Target Classes

Exploratory Data Analysis (EDA)

What Was Examined

- Reviewed dataset shape, column types, and descriptive statistics
- Checked for missing values, duplicates, and class distribution imbalances
- Visualized data using histograms, boxplots, and a correlation heatmap to understand feature relationships



Data Preprocessing

1

Clean

Removed unnecessary empty columns from the raw dataset

2

Encode

Diagnosis labels encoded: **Benign** = **0**, **Malignant** = **1**

3

Scale

All features standardized using **StandardScaler** for uniform range

4

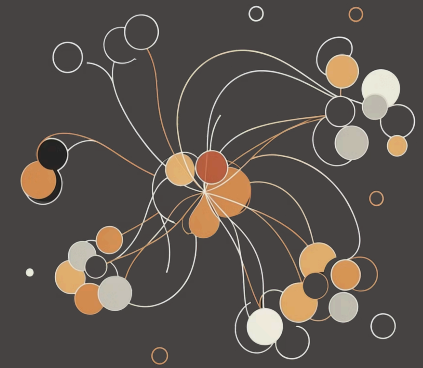
Split

80% Training (455 samples) / **20% Testing** (114 samples)

How KNN Works

- Classifies a new sample based on the **majority class** among its K nearest neighbors in the training set
- Uses **Euclidean distance** to measure similarity between data points
- Model trained with **n_neighbors = 5** using Scikit-learn's KNeighborsClassifier

❏ KNN is a lazy learner — it memorizes the training data and classifies new points at prediction time rather than building an explicit model.



Model Evaluation Results

95.61%

Test Accuracy

Strong predictive performance on 114
unseen test samples

455

Training Samples

80% of the full dataset used for model
training

114

Testing Samples

20% held out for unbiased model
evaluation

Split	Samples	Accuracy
Training	455	—
Testing	114	95.61%

The classification report confirms strong precision, recall, and F1-scores across both benign and malignant classes. The confusion matrix validates robust performance with minimal misclassifications.

Advantages & Limitations of KNN



Advantages

- Simple and intuitive — easy to understand and implement
- Non-parametric: makes no assumptions about data distribution
- No explicit training phase required
- Performs well when features are properly standardized

Limitations

- Sensitive to the choice of K value — wrong K can reduce accuracy
- Requires careful feature scaling for meaningful distance calculation
- High computational cost on large datasets at prediction time
- Performance degrades with noisy or irrelevant features

Conclusion & Future Work

Key Outcome

The KNN classifier achieved **95.61% accuracy** on the Wisconsin Breast Cancer Diagnostic dataset — demonstrating that a well-tuned, standardized KNN model can serve as a reliable tool for early breast cancer diagnosis support.

Future Improvements

- Tune the optimal K using the **Elbow Method**
- Apply **cross-validation** for more robust performance estimation
- Compare results against **SVM, Random Forest, and Logistic Regression**

